Class 12 Mathematics

Chapter 4: Determinants

Practice Set A: step-wise solutions

20 questions **54** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** PYQ-style 1 mark

If $\begin{vmatrix} x+1 & x-1 \\ x-3 & x+2 \end{vmatrix} = \begin{vmatrix} 4 & -1 \\ 1 & 3 \end{vmatrix}$, then x is equal to:

(A) $\dfrac{12}{7}$

(B) 2

(C) -2

(D) $\dfrac{8}{7}$

Formula / idea: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

1 Expand the LHS.
$(x+1)(x+2) - (x-1)(x-3) = (x^2 + 3x + 2) - (x^2 - 4x + 3) = 7x - 1$

2 Expand the RHS.
$4(3) - (-1)(1) = 12 + 1 = 13$

3 Equate and solve. ✓ 1
$7x - 1 = 13 \Rightarrow x = 2$

4 Hence option (B).

Answer **(B)** 2

Watch out: Writing the RHS as $12 - 1 = 11$; here $bc = (-1)(1) = -1$, so $ad - bc = 12 + 1$.

2 MCQ CBSE 2023 1 mark

Let A be a 3×3 matrix such that $|\text{adj } A| = 64$. Then $|A|$ is equal to:

(A) 8 only **(B)** -8 only

(C) 64 **(D)** 8 **or** -8

Formula / idea: For a square matrix of order n: $|\text{adj } A| = |A|^{n-1}$.

1 Use the result with $n = 3$.
$|\text{adj } A| = |A|^{3-1} = |A|^2$

2 Put $|\text{adj } A| = 64$ and solve. ✓ 1
$|A|^2 = 64 \Rightarrow |A| = \pm 8$

3 Both values are possible. Option (D).

Answer **(D)** 8 **or** -8

Watch out: Taking only the positive square root; $|A|$ can be negative.

3 MCQ CBSE 2023 1 mark

If $|A| = 2$, where A is a 2×2 matrix, then $|4A^{-1}|$ equals:

(A) 4

(B) 2

(C) 8

(D) $\dfrac{1}{32}$

Formula / idea: $|kB| = k^n|B|$ for order n, and $|A^{-1}| = \dfrac{1}{|A|}$.

1 Find $|A^{-1}|$.
$|A^{-1}| = \dfrac{1}{|A|} = \dfrac{1}{2}$

2 Use $|kB| = k^n|B|$ with $k = 4$, $n = 2$. ✓ 1
$|4A^{-1}| = 4^2 \cdot \dfrac{1}{2} = 16 \cdot \dfrac{1}{2} = 8$

3 Hence option (C).

Answer **(C)** 8

Watch out: Writing $|4A^{-1}| = 4 \cdot \frac{1}{2} = 2$; the factor 4 comes out as 4^2 for a 2×2 matrix.

4 MCQ CBSE SQP 2024-25 1 mark

If the points (x_1, y_1), (x_2, y_2) and $(x_1 + x_2,\ y_1 + y_2)$ are collinear, then x_1y_2 is equal to:

(A) x_2y_1 **(B)** x_1y_1

(C) x_2y_2 **(D)** x_1x_2

Formula / idea: Three points are collinear when the area of the triangle they form is 0.

1 Collinear, so the area determinant is zero.

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_1 + x_2 & y_1 + y_2 & 1 \end{vmatrix} = 0$$

2 Expand along R_1.

$$x_1(y_2 - y_1 - y_2) - y_1(x_2 - x_1 - x_2) + (x_2y_1 + x_2y_2 - x_1y_2 - x_2y_2) = 0$$

3 Simplify. ✓ 1

$$-x_1y_1 + x_1y_1 + x_2y_1 - x_1y_2 = 0 \ \Rightarrow\ x_1y_2 = x_2y_1$$

4 Hence option (A).

Answer **(A)** x_2y_1

5 MCQ PYQ-style 1 mark

If A_{ij} is the cofactor of the element a_{ij} of the determinant $\begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$, then the value of $a_{32} \cdot A_{32}$ is:

(A) 22 **(B)** -22

(C) 11 **(D)** -11

Formula / idea: $A_{ij} = (-1)^{i+j}M_{ij}$, where M_{ij} is the minor of a_{ij}.

1 Read the element in row 3, column 2.

$$a_{32} = 2$$

2 Minor: delete row 3 and column 2.

$$M_{32} = \begin{vmatrix} 5 & 8 \\ 2 & 1 \end{vmatrix} = 5 - 16 = -11$$

3 Cofactor: attach the sign $(-1)^{3+2}$.

$$A_{32} = (-1)^5(-11) = 11$$

4 Multiply. ✓ 1

$$a_{32} \cdot A_{32} = 2 \times 11 = 22$$

5 Hence option (A).

Answer **(A)** 22

Watch out: Using the minor -11 in place of the cofactor and getting -22.

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): If A is a square matrix of order 3 and $|A| = 3$, then $|\text{adj } A| = 27$. **Reason (R):** For a square matrix A of order n, $|\text{adj } A| = |A|^{n-1}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) **Assertion (A) is false, but Reason (R) is true.**

Formula / idea: $|\text{adj } A| = |A|^{n-1}$ for a square matrix of order n.

1 Check A using the result with $n = 3$.
$|\text{adj } A| = |A|^{3-1} = 3^2 = 9 \neq 27$

2 So **A is false.**

3 R is the standard result for order n. **R is true.**

4 A false, R true. Hence option (D). ✓ 1

Answer **(D) Assertion (A) is false, but Reason (R) is true.**

Watch out: Using the power n instead of $n-1$, which gives $3^3 = 27$.

7 **Short answer** PYQ-style 2 marks

Find the values of k if the area of the triangle with vertices $(2, -6)$, $(5, 4)$ and $(k, 4)$ is 35 sq units.

Formula / idea: Area $= \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$

1 Write the area using determinants. ✓ ½

$$\Delta = \frac{1}{2}\begin{vmatrix} 2 & -6 & 1 \\ 5 & 4 & 1 \\ k & 4 & 1 \end{vmatrix} = \pm 35$$

Area is positive, so take both signs.

2	Expand along R_1. $\frac{1}{2}\left[2(4-4)+6(5-k)+1(20-4k)\right]=\frac{1}{2}(50-10k)=25-5k$	✓ ½
3	Equate to ± 35. $25-5k=35 \quad \text{or} \quad 25-5k=-35$	✓ ½
4	Solve each equation. $k=-2 \quad \text{or} \quad k=12$	✓ ½
5	Hence $k=12$ or $k=-2$.	

Answer $k=12$ **or** $k=-2$

Watch out: Taking only $+35$ and missing the second value $k=12$.

Level 2

17 marks

8 **Short answer** PYQ-style 2 marks

Find the adjoint and the inverse of the matrix $A=\begin{bmatrix}2 & -2\\ 4 & 3\end{bmatrix}$.

Formula / idea: $A^{-1}=\frac{1}{|A|}\operatorname{adj} A$, when $|A|\neq 0$.

1	Find $\lvert A\rvert$. $\lvert A\rvert=2(3)-(-2)(4)=6+8=14\neq 0$ So A^{-1} exists.	✓ ½
2	For a 2×2 matrix: swap the diagonal, change signs of the other two. $\operatorname{adj} A=\begin{bmatrix}3 & 2\\ -4 & 2\end{bmatrix}$	✓ 1
3	Divide by $\lvert A\rvert$. $A^{-1}=\frac{1}{|A|}\operatorname{adj} A=\frac{1}{14}\begin{bmatrix}3 & 2\\ -4 & 2\end{bmatrix}$	✓ ½

Answer $\operatorname{adj} A=\begin{bmatrix}3 & 2\\ -4 & 2\end{bmatrix}$, $A^{-1}=\frac{1}{14}\begin{bmatrix}3 & 2\\ -4 & 2\end{bmatrix}$

Watch out: Changing the signs of the diagonal elements instead of the off-diagonal ones.

9 **Short answer** PYQ-style 3 marks

For the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$, find the numbers a and b such that $A^2 + aA + bI = O$. Hence find A^{-1}.

1 Find A^2. ✓ 1

$$A^2 = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 8 \\ 4 & 3 \end{bmatrix}$$

2 Substitute in $A^2 + aA + bI = O$. ✓ ½

$$\begin{bmatrix} 11+3a+b & 8+2a \\ 4+a & 3+a+b \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

3 Equate the $(2,1)$ elements, then the $(1,1)$ elements. ✓ ½

$4 + a = 0 \Rightarrow a = -4; \qquad 11 - 12 + b = 0 \Rightarrow b = 1$

4 So $A^2 - 4A + I = O$. Also $|A| = 3 - 2 = 1 \neq 0$, so A^{-1} exists.

5 Multiply both sides by A^{-1}. ✓ ½

$A - 4I + A^{-1} = O \;\Rightarrow\; A^{-1} = 4I - A$

6 Compute $4I - A$. ✓ ½

$$A^{-1} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$$

Answer $a = -4, b = 1$; $A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$

Watch out: Writing $A^{-1}I$ as I while multiplying through; it is A^{-1}.

10 **Short answer** PYQ-style 2 marks

Using determinants, find the equation of the line joining $A(1,3)$ and $B(0,0)$. Hence find k if $D(k,0)$ is a point such that the area of $\triangle ABD$ is 3 sq units.

Formula / idea: $P(x,y)$ lies on line AB when the area of $\triangle PAB$ is 0.

1 Let $P(x,y)$ be any point on AB. Area of $\triangle PAB = 0$.

$$\begin{vmatrix} x & y & 1 \\ 1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 0$$

2 Expand along R_1. ✓ 1

$x(3-0) - y(1-0) + 1(0-0) = 0 \;\Rightarrow\; y = 3x$

3 Area of $\triangle ABD$ using determinants.

$$\frac{1}{2}\begin{vmatrix} 1 & 3 & 1 \\ 0 & 0 & 1 \\ k & 0 & 1 \end{vmatrix} = \pm 3$$

4 Expand along R_1.

$$\frac{1}{2}\left[1(0-0)-3(0-k)+1(0-0)\right]=\frac{3k}{2}=\pm 3$$

✓ ½

5 Solve.

$k=\pm 2$

✓ ½

Answer $y=3x;\ k=\pm 2$

Watch out: Forgetting the $\pm$ sign for the area and giving only $k=2$.

11 **Short answer** PYQ-style 3 marks

Solve the following system of equations by the matrix method: $2x+5y=1,\ 3x+2y=7$.

Formula / idea: $AX=B\Rightarrow X=A^{-1}B$, when $|A|\neq 0$.

1 Write the system as $AX=B$.

$$\begin{bmatrix}2 & 5\\ 3 & 2\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}1\\ 7\end{bmatrix}$$

✓ ½

2 Find $|A|$.

$|A|=4-15=-11\neq 0$

So a unique solution exists.

✓ ½

3 Find $\operatorname{adj} A$.

$$\operatorname{adj} A=\begin{bmatrix}2 & -5\\ -3 & 2\end{bmatrix}$$

✓ ½

4 Find A^{-1}.

$$A^{-1}=-\frac{1}{11}\begin{bmatrix}2 & -5\\ -3 & 2\end{bmatrix}$$

✓ ½

5 Find $X=A^{-1}B$.

$$X=-\frac{1}{11}\begin{bmatrix}2-35\\ -3+14\end{bmatrix}=-\frac{1}{11}\begin{bmatrix}-33\\ 11\end{bmatrix}=\begin{bmatrix}3\\ -1\end{bmatrix}$$

✓ ½

6 Hence $x=3,\ y=-1$.

✓ ½

Answer $x=3,\ y=-1$

Watch out: Writing $X=BA^{-1}$; the order must be $A^{-1}B$.

12 **Short answer** Board-pattern 2 marks

Examine the consistency of the system of equations: $2x - 3y = 5,\ 6x - 9y = 15$.

Formula / idea: If $|A| = 0$ and $(\text{adj } A)B = O$, the system may be consistent or inconsistent; check the equations.

1	Write as $AX = B$ and find $\|A\|$. $A = \begin{bmatrix} 2 & -3 \\ 6 & -9 \end{bmatrix},\ B = \begin{bmatrix} 5 \\ 15 \end{bmatrix},\ \|A\| = -18 + 18 = 0$	✓ ½
2	Find $\text{adj } A$. $\text{adj } A = \begin{bmatrix} -9 & 3 \\ -6 & 2 \end{bmatrix}$	✓ ½
3	Find $(\text{adj } A)B$. $(\text{adj } A)B = \begin{bmatrix} -45 + 45 \\ -30 + 30 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = O$	✓ ½
4	The second equation is 3 times the first, so both are the same line.	
5	Hence the system is consistent, with infinitely many solutions.	✓ ½

Answer **Consistent, with infinitely many solutions.**

Watch out: Calling the system inconsistent just because $|A| = 0$.

13 **Short answer** Board-pattern 2 marks

If $A = \begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix}$, show that $A^{-1} = A$.

1	Find $\|A\|$. $\|A\| = 3(-3) - (-4)(2) = -9 + 8 = -1 \neq 0$	✓ ½
2	Find $\text{adj } A$. $\text{adj } A = \begin{bmatrix} -3 & 4 \\ -2 & 3 \end{bmatrix}$	✓ ½
3	Use $A^{-1} = \dfrac{1}{\|A\|}\,\text{adj } A$. $A^{-1} = \dfrac{1}{-1}\begin{bmatrix} -3 & 4 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix}$	✓ ½
4	This is the matrix A. Hence $A^{-1} = A$.	✓ ½

Answer $A^{-1} = \begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix} = A$

Watch out: Dropping the minus sign of $|A| = -1$ when dividing.

14 **Short answer** PYQ-style 3 marks

If $A = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 8 \\ 7 & 9 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$.

1 Find AB. ✓ ½

$$AB = \begin{bmatrix} 18+49 & 24+63 \\ 12+35 & 16+45 \end{bmatrix} = \begin{bmatrix} 67 & 87 \\ 47 & 61 \end{bmatrix}$$

2 Find $|AB|$.

$$|AB| = 67 \times 61 - 87 \times 47 = 4087 - 4089 = -2$$

3 Find $(AB)^{-1}$. ✓ 1

$$(AB)^{-1} = -\frac{1}{2}\begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix}$$

4 Find A^{-1}; here $|A| = 15 - 14 = 1$. ✓ ½

$$A^{-1} = \begin{bmatrix} 5 & -7 \\ -2 & 3 \end{bmatrix}$$

5 Find B^{-1}; here $|B| = 54 - 56 = -2$. ✓ ½

$$B^{-1} = -\frac{1}{2}\begin{bmatrix} 9 & -8 \\ -7 & 6 \end{bmatrix}$$

6 Find $B^{-1}A^{-1}$.

$$B^{-1}A^{-1} = -\frac{1}{2}\begin{bmatrix} 45+16 & -63-24 \\ -35-12 & 49+18 \end{bmatrix} = -\frac{1}{2}\begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix}$$

7 Both sides are equal. Hence $(AB)^{-1} = B^{-1}A^{-1}$ is verified. ✓ ½

Answer $(AB)^{-1} = B^{-1}A^{-1} = -\frac{1}{2}\begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix}$. **Verified.**

Watch out: Multiplying in the order $A^{-1}B^{-1}$; the reversal law needs $B^{-1}A^{-1}$.

Level 3

29 marks

15 **Long answer** CBSE SQP 2024-25 5 marks

The equation of the path traversed by the ball headed by a footballer is $y = ax^2 + bx + c$ (where $0 \le x \le 14$ and $a, b, c \in \mathbb{R}$, $a \neq 0$) with respect to an XY-coordinate system in the vertical plane. The ball passes through the points $(2, 15)$, $(4, 25)$ and $(14, 15)$. Determine the values of a, b and c by solving the system of linear equations in a, b and c, using the matrix method. Also find the equation of the path traversed by the ball.

Formula / idea: $AX = B \Rightarrow X = A^{-1}B = \dfrac{1}{|A|}(\text{adj } A)B.$

1 Put each point in $y = ax^2 + bx + c$. ✓ 1

$4a + 2b + c = 15, \quad 16a + 4b + c = 25, \quad 196a + 14b + c = 15$

2 Write as $AX = B$.

$$\begin{bmatrix} 4 & 2 & 1 \\ 16 & 4 & 1 \\ 196 & 14 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 15 \\ 25 \\ 15 \end{bmatrix}$$

3 Find $|A|$ by expanding along R_1. ✓ ½

$|A| = 4(4 - 14) - 2(16 - 196) + 1(224 - 784) = -40 + 360 - 560 = -240 \neq 0$

4 Cofactors of row 1. ✓ ½

$A_{11} = -10, \quad A_{12} = 180, \quad A_{13} = -560$

5 Cofactors of row 2. ✓ ½

$A_{21} = 12, \quad A_{22} = -192, \quad A_{23} = 336$

6 Cofactors of row 3. ✓ ½

$A_{31} = -2, \quad A_{32} = 12, \quad A_{33} = -16$

7 $\text{adj } A$ is the transpose of the cofactor matrix. ✓ ½

$$\text{adj } A = \begin{bmatrix} -10 & 12 & -2 \\ 180 & -192 & 12 \\ -560 & 336 & -16 \end{bmatrix}$$

8 Write A^{-1}.

$$A^{-1} = -\frac{1}{240}\begin{bmatrix} -10 & 12 & -2 \\ 180 & -192 & 12 \\ -560 & 336 & -16 \end{bmatrix}$$

9 Find $X = A^{-1}B$. ✓ ½

$$X = -\frac{1}{240}\begin{bmatrix} -150 + 300 - 30 \\ 2700 - 4800 + 180 \\ -8400 + 8400 - 240 \end{bmatrix} = -\frac{1}{240}\begin{bmatrix} 120 \\ -1920 \\ -240 \end{bmatrix}$$

10 Simplify. ✓ ½

$a = -\dfrac{1}{2}, \quad b = 8, \quad c = 1$

11 Hence the path of the ball is $y = -\frac{1}{2}x^2 + 8x + 1$. ✓ ½

Answer $a = -\frac{1}{2}$, $b = 8$, $c = 1$; **path:** $y = -\frac{1}{2}x^2 + 8x + 1$

Watch out: Writing (x, y) as (a, b); the unknowns are a, b, c and each point gives one equation.

16 **Long answer** CBSE SQP 2025-26 5 marks

For two matrices $A = \begin{bmatrix} 3 & -6 & -1 \\ 2 & -5 & -1 \\ -2 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 & -1 \\ 0 & -1 & -1 \\ 2 & 0 & 3 \end{bmatrix}$, find the product AB and hence solve the system of equations: $3x - 6y - z = 3$, $2x - 5y - z + 2 = 0$, $-2x + 4y + z = 5$.

Formula / idea: If $AB = I$, then $A^{-1} = B$.

1 Row 1 of AB. ✓ ½
$(3 + 0 - 2,\ -6 + 6 + 0,\ -3 + 6 - 3) = (1,\ 0,\ 0)$

2 Row 2 of AB. ✓ ½
$(2 + 0 - 2,\ -4 + 5 + 0,\ -2 + 5 - 3) = (0,\ 1,\ 0)$

3 Row 3 of AB. ✓ ½
$(-2 + 0 + 2,\ 4 - 4 + 0,\ 2 - 4 + 3) = (0,\ 0,\ 1)$

4 So AB is the identity matrix. ✓ ½
$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$

5 Hence $A^{-1} = B$. ✓ 1
$AB = I \Rightarrow A^{-1} = B$

6 Write the system as $AX = C$; note $2x - 5y - z = -2$. ✓ ½
$\begin{bmatrix} 3 & -6 & -1 \\ 2 & -5 & -1 \\ -2 & 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}$

7 $X = A^{-1}C = BC$. ✓ 1
$X = \begin{bmatrix} 1 & -2 & -1 \\ 0 & -1 & -1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 + 4 - 5 \\ 0 + 2 - 5 \\ 6 + 0 + 15 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 21 \end{bmatrix}$

8 Hence $x = 2$, $y = -3$, $z = 21$. ✓ ½

Answer $AB = I$; $x = 2$, $y = -3$, $z = 21$

Watch out: Using $+2$ on the right of the second equation; move it across first: $2x - 5y - z = -2$.

17 **Long answer** CBSE 2023 5 marks

Solve the following system of equations by the matrix method: $x + 2y + 3z = 6$, $2x - y + z = 2$, $3x + 2y - 2z = 3$.

Formula / idea: $AX = B \Rightarrow X = A^{-1}B$, where $A^{-1} = \dfrac{1}{|A|}\operatorname{adj} A$.

1	Write as $AX = B$. $\begin{bmatrix}1 & 2 & 3\\ 2 & -1 & 1\\ 3 & 2 & -2\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}6\\ 2\\ 3\end{bmatrix}$	✓ ½
2	Find $\lvert A\rvert$ by expanding along R_1. $\lvert A\rvert = 1(2-2) - 2(-4-3) + 3(4+3) = 0 + 14 + 21 = 35 \neq 0$	✓ 1
3	Cofactors of row 1. $A_{11} = 0, \quad A_{12} = 7, \quad A_{13} = 7$	✓ ½
4	Cofactors of row 2. $A_{21} = 10, \quad A_{22} = -11, \quad A_{23} = 4$	✓ ½
5	Cofactors of row 3. $A_{31} = 5, \quad A_{32} = 5, \quad A_{33} = -5$	✓ ½
6	$\operatorname{adj} A$ is the transpose of the cofactor matrix. $\operatorname{adj} A = \begin{bmatrix}0 & 10 & 5\\ 7 & -11 & 5\\ 7 & 4 & -5\end{bmatrix}$	✓ ½
7	Write A^{-1}. $A^{-1} = \dfrac{1}{35}\begin{bmatrix}0 & 10 & 5\\ 7 & -11 & 5\\ 7 & 4 & -5\end{bmatrix}$	✓ ½
8	Find $X = A^{-1}B$. $X = \dfrac{1}{35}\begin{bmatrix}0+20+15\\ 42-22+15\\ 42+8-15\end{bmatrix} = \dfrac{1}{35}\begin{bmatrix}35\\ 35\\ 35\end{bmatrix} = \begin{bmatrix}1\\ 1\\ 1\end{bmatrix}$	✓ ½
9	Hence $x = 1, y = 1, z = 1$.	✓ ½

Answer $x = 1, y = 1, z = 1$

Watch out: Forgetting to transpose the cofactor matrix when writing $\operatorname{adj} A$.

18 **Long answer** PYQ-style 5 marks

Gautam buys 5 pens, 3 bags and 1 instrument box and pays ₹160. From the same shop, Vikram buys 2 pens, 1 bag and 3 instrument boxes and pays ₹190. Ankur buys 1 pen, 2 bags and 4 instrument boxes and pays ₹250. Using the matrix method, find the cost of one pen, one bag and one instrument box.

Formula / idea: $AX = B \Rightarrow X = A^{-1}B$, where $A^{-1} = \dfrac{1}{|A|}\operatorname{adj} A$.

1	Let one pen, bag, box cost ₹x, ₹y, ₹z. $5x + 3y + z = 160, \quad 2x + y + 3z = 190, \quad x + 2y + 4z = 250$	✓ ½
2	Write as $AX = B$. $\begin{bmatrix}5 & 3 & 1\\ 2 & 1 & 3\\ 1 & 2 & 4\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}160\\ 190\\ 250\end{bmatrix}$	✓ ½
3	Find $\lvert A\rvert$ by expanding along R_1. $\lvert A\rvert = 5(4-6) - 3(8-3) + 1(4-1) = -10 - 15 + 3 = -22 \neq 0$	✓ ½
4	Cofactors of row 1. $A_{11} = -2, \quad A_{12} = -5, \quad A_{13} = 3$	✓ ½
5	Cofactors of row 2. $A_{21} = -10, \quad A_{22} = 19, \quad A_{23} = -7$	✓ ½
6	Cofactors of row 3. $A_{31} = 8, \quad A_{32} = -13, \quad A_{33} = -1$	✓ ½
7	$\operatorname{adj} A$ is the transpose of the cofactor matrix. $\operatorname{adj} A = \begin{bmatrix}-2 & -10 & 8\\ -5 & 19 & -13\\ 3 & -7 & -1\end{bmatrix}$	✓ ½
8	Write A^{-1}. $A^{-1} = -\dfrac{1}{22}\begin{bmatrix}-2 & -10 & 8\\ -5 & 19 & -13\\ 3 & -7 & -1\end{bmatrix}$	✓ ½
9	Find $X = A^{-1}B$. $X = -\dfrac{1}{22}\begin{bmatrix}-320 - 1900 + 2000\\ -800 + 3610 - 3250\\ 480 - 1330 - 250\end{bmatrix} = -\dfrac{1}{22}\begin{bmatrix}-220\\ -440\\ -1100\end{bmatrix} = \begin{bmatrix}10\\ 20\\ 50\end{bmatrix}$	✓ ½
10	Hence a pen costs ₹10, a bag ₹20 and an instrument box ₹50.	✓ ½

Answer **Pen ₹10, bag ₹20, instrument box ₹50**

Watch out: Mixing up rows: each person's purchase is one equation, so it forms one row of A.

19 **Long answer** CBSE 2025 Compartment **5 marks**

If $A = \begin{bmatrix} 3 & 2 & 1 \\ 4 & -1 & 2 \\ 7 & 3 & -3 \end{bmatrix}$, find A^{-1}. Using A^{-1}, solve the system of equations: $3x + 4y + 7z = 14$, $2x - y + 3z = 4$, $x + 2y - 3z = 0$.

Formula / idea: The coefficient matrix here is A', and $(A')^{-1} = (A^{-1})'$.

1 Find $|A|$ by expanding along R_1. ✓ ½

$|A| = 3(3-6) - 2(-12-14) + 1(12+7) = -9 + 52 + 19 = 62 \neq 0$

2 Cofactors of row 1. ✓ ½

$A_{11} = -3, \quad A_{12} = 26, \quad A_{13} = 19$

3 Cofactors of row 2. ✓ ½

$A_{21} = 9, \quad A_{22} = -16, \quad A_{23} = 5$

4 Cofactors of row 3. ✓ ½

$A_{31} = 5, \quad A_{32} = -2, \quad A_{33} = -11$

5 $\text{adj}\, A$ is the transpose of the cofactor matrix. ✓ ½

$\text{adj}\, A = \begin{bmatrix} -3 & 9 & 5 \\ 26 & -16 & -2 \\ 19 & 5 & -11 \end{bmatrix}$

6 Write A^{-1}. ✓ ½

$A^{-1} = \frac{1}{62} \begin{bmatrix} -3 & 9 & 5 \\ 26 & -16 & -2 \\ 19 & 5 & -11 \end{bmatrix}$

7 The columns of A are the rows of the system, so it reads $A'X = B$. ✓ ½

$A'X = B, \quad B = \begin{bmatrix} 14 \\ 4 \\ 0 \end{bmatrix}$

8 So $X = (A')^{-1}B = (A^{-1})'B$. ✓ ½

$X = \frac{1}{62} \begin{bmatrix} -3 & 26 & 19 \\ 9 & -16 & 5 \\ 5 & -2 & -11 \end{bmatrix} \begin{bmatrix} 14 \\ 4 \\ 0 \end{bmatrix}$

9 Multiply. ✓ ½

$X = \frac{1}{62} \begin{bmatrix} -42 + 104 + 0 \\ 126 - 64 + 0 \\ 70 - 8 + 0 \end{bmatrix} = \frac{1}{62} \begin{bmatrix} 62 \\ 62 \\ 62 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

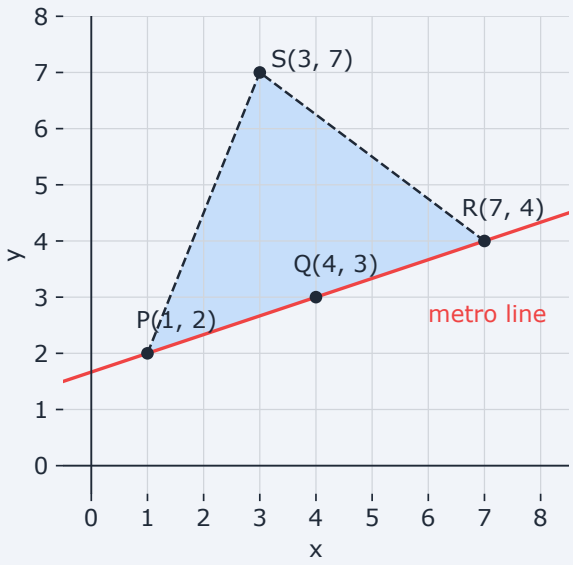
10 Hence $x = 1, y = 1, z = 1$. ✓ ½

Answer $A^{-1} = \frac{1}{62}\begin{bmatrix} -3 & 9 & 5 \\ 26 & -16 & -2 \\ 19 & 5 & -11 \end{bmatrix}$; $x = 1, y = 1, z = 1$

Watch out: Using $X = A^{-1}B$ directly; the coefficient matrix is A', not A.

20 **Case study** Board-pattern 4 marks

Case Study: A city planner marks three metro stations on a map grid at $P(1, 2)$, $Q(4, 3)$ and $R(7, 4)$. A new park is planned at $S(3, 7)$ (1 unit $= 100$ m).



(i) Using determinants, show that the stations P, Q and R lie on one straight line. 1 mark

(ii) Using determinants, find the area of the triangular plot with corners P, R and S. 1 mark

(iii) Using determinants, find the equation of the line through P and R. 2 marks

Formula / idea: Area $= \frac{1}{2}\left|\begin{matrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{matrix}\right|$; area 0 means collinear.

(i) 1 mark

1 Find the area determinant for P, Q, R. ✓ ½

$$\begin{vmatrix} 1 & 2 & 1 \\ 4 & 3 & 1 \\ 7 & 4 & 1 \end{vmatrix} = 1(3-4) - 2(4-7) + 1(16-21) = -1 + 6 - 5 = 0$$

2 Area is 0, so P, Q, R lie on one straight line. ✓ ½

Answer (i) **Determinant $= 0$, so P, Q, R are collinear.**

(ii) 1 mark

1 Write the area of $\triangle PRS$.

$$\Delta = \frac{1}{2}\begin{vmatrix} 1 & 2 & 1 \\ 7 & 4 & 1 \\ 3 & 7 & 1 \end{vmatrix} = \frac{1}{2}\left[1(4-7) - 2(7-3) + 1(49-12)\right]$$

✓ ½

2 Simplify.

$$\Delta = \frac{1}{2}(-3 - 8 + 37) = \frac{26}{2} = 13 \text{ sq units}$$

✓ ½

Answer (ii) 13 **sq units**

(iii) 2 marks

1 Let (x, y) be any point on PR. The area with P and R is 0.

$$\begin{vmatrix} x & y & 1 \\ 1 & 2 & 1 \\ 7 & 4 & 1 \end{vmatrix} = 0$$

✓ ½

2 Expand along R_1.

$$x(2-4) - y(1-7) + 1(4-14) = 0$$

✓ ½

3 Simplify.

$$-2x + 6y - 10 = 0$$

✓ ½

4 Divide by -2. Required line: $x - 3y + 5 = 0$.

✓ ½

Answer (iii) $x - 3y + 5 = 0$

Watch out: Leaving the area negative; area is the absolute value of the determinant expression.